

6-Month Cybersecurity Curriculum for Beginners

Duration: 24 weeks (6 months)

Level: Beginner → Intermediate

Includes: Theory + Practical + Projects + Assessment

Month 1 – Cybersecurity Foundations

Week 1: Introduction to Cybersecurity

- What is cybersecurity?
- Importance of cybersecurity
- Real-world cyber attack examples
- CIA Triad (Confidentiality, Integrity, Availability)

Week 2: Types of Cyber Attacks

- Malware (Virus, Worm, Trojan, Ransomware)
- Phishing and Social Engineering
- DoS/DDoS
- Password attacks

Week 3: Essential Cyber Hygiene

- Strong passwords
- 2FA/MFA
- Updating systems
- Safe downloading
- Recognizing suspicious links

Week 4: Computer & Mobile Security Basics

- Antivirus & firewall
 - Safe browser settings
 - App permissions
 - Removing malware
 - Activity: Students secure their own device
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Month 2 – Networking Basics & Security

Week 5: Computer Networks – Basics

- What is a network?
- LAN, WAN, Internet
- IP address, MAC address
- Ports & protocols

Week 6: Network Devices

- Router, switch, modem
- Access points
- Firewalls (simple explanation)

Week 7: Network Security

- Types of network attacks
- Secure Wi-Fi configuration
- Encryption (WPA2/WPA3)
- VPN basics

Week 8: Hands-On Activities

- Checking open ports
- Packet capture demo (Wireshark — beginner level)

- Securing a Wi-Fi network
 - Activity worksheets
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Month 3 – System & Application Security

Week 9: Operating System Security

- Windows security features
- Linux basics
- User accounts & permissions
- Updates & patches

Week 10: Application Security

- Browser security
- Mobile app security
- Software vulnerabilities
- OWASP basics (for beginners)

Week 11: Cloud Security Basics

- What is cloud?
- Cloud threats
- Safe usage of Google Drive, OneDrive, etc.

Week 12: Email & Web Security

- Identifying phishing emails
 - HTTPS, SSL/TLS
 - Safe browsing habits
 - Activity: Spot phishing email challenge
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Month 4 – Introduction to Ethical Hacking

Week 13: Introduction to Ethical Hacking

- White hat vs black hat
- Bug bounty basics
- Legal & ethical guidelines

Week 14: Reconnaissance (Information Gathering)

- What is footprinting?
- Google dorking basics
- Whois, NSlookup demo

Week 15: Scanning & Enumeration

- What are ports?
- Simple Nmap scan (safe practice)
- Identifying open services

Week 16: Vulnerability Basics

- What is a vulnerability?
- Common vulnerabilities
- CVE database introduction
- Activity: Identify vulnerabilities in simple examples

Month 5 – Practical Tools & Techniques

Week 17: Password Cracking – Basics

- Hashing explained simply
- Brute force vs dictionary attack

- Demo: Hash identifier & cracking with safe tools (only on sample files)

Week 18: Wireshark Basics

- How network packets work
- Capture, filter, analyze
- Activity: Students identify packet types

Week 19: Web Application Security Basics

- SQL Injection (very simple explanation)
- XSS basics
- Broken authentication (simple cases)

Week 20: Secure Coding Basics

- Input validation
- Sanitization
- Common coding mistakes leading to vulnerabilities
- Activity: Fix insecure code examples

Month 6 – Projects, Labs & Career Skills

Week 21: Digital Forensics Basics

- What is digital forensics?
- File recovery basics
- Log analysis basics

Week 22: Cyber Laws & Ethics

- Local cybercrime laws (India IT Act basics if needed)
- Cyber ethics

- Privacy rights

Week 23: Mini Projects

Students choose a project such as:

- Create a “Phishing Awareness” campaign
- Create a strong password policy for a school
- Secure Wi-Fi setup demonstration
- Web security poster
- Basic vulnerability report of a dummy website

Week 24: Final Assessment

- Practical exam
- Theory exam
- Project presentation
- Feedback session
- Certification